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Computer Vision
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① Computer Vision enable computers to interpret with AI techniques and algorithms used in it to understand visual information.

a) Objective of Computer Vision:

- * It interprets the visual informations.
- * Scene analysis.
- * Image recognition.

b) Real-World Applications of Computer Visions:

- * Facial Recognition.
- * Self-driving Cars.

c) Key Features:

- * Feature extraction
- * Pattern recognition
- * Object Detection

a) Feature extraction:

Feature extraction is the process of identifying important characteristics of an image.

b) The object recognition improve Computer Vision system by object detection and recognition.

4) Levels of image processing used in Computer Visions are:

- * Low Level Image
- * Mid Level Image
- * High Level Image

a) Low-level Image Processing:

The Computer Vision extract the image and noise filters and enhancing the images.

b) Mid level image Processing:

The image interprets and sharpens the end of the image or curves it etc.

High-level image Processing:

Here the image process the object recognition and image extraction.

4) The Digital Image represents quality and brightness of the Image and the matrix of pixel.

a) Pixel:

* Pixel is the smallest element of a digital image that stores colour or intensity.

* Values from 0 to 255

The image resolution affect the image quality by providing more image details in better quality and clarity.

6) Sampling and quantization is the resolution of an image, while determine the intensity level.

a) Sampling :

It is the process of converting a continuous image into discrete pixels.

b) The Purpose of Quantization :

Quantization converts continuous intensity values into discrete values.

7) Image formation model the process of capturing Images.

a) Converting the 3D images into 2D images is called perspective projection.

b) Illumination provides the light required to make objects visible for image capture.

6) ~~Feature extraction is the process of identifying the~~

8) a) Image processing is the manipulation and enhancement of digital Image.

b) Image Processing:

- * Focuses on image enhancement and restoration.

- * Produces improved images.

Computer Vision:

- * Focuses on understanding and interpreting images.

- * Produces decisions.

1) Healthcare:

Medical image analysis for disease detection.

2) Application in autonomous vehicle:

lane detection and obstacle recognition

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c) Pixels and image is composed of pixels which contain image information.

a) Pixel Important :

Every digital image is composed of pixel, which contain image information.

b) Sampling and image resolution influence image analysis is more detailed about image and improves analysis accuracy.